

Transformers for Mental Health: Comparative Analysis of BERT, RoBERTa, DistilBERT, and ELECTRA in Detecting Depression from Reddit Posts

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Abstract— This study presents a comparative analysis of making them particularly suitable for mental health classification transformer-based language models—BERT, RoBERTa, tasks. By utilizing embeddings generated from these models and DistilBERT, and Electra—for the classification of mental health indicators in social media posts. Using a dataset of training a Logistic Regression classifier, we aim to evaluate their effectiveness in accurately identifying signs of depression. This work not only contributes to the development of robust NLP methods for mental health detection but also provides a comparative analysis of multiple transformer models in this critical domain. Each model's tokenizer was initialized to preprocess posts with standardized token length, truncation, and padding. The generated embeddings were averaged across the last hidden layer and fed into a Logistic Regression classifier, accounting for class imbalance through calculated weights. Performance evaluation was conducted using accuracy, precision, recall, and F1-score metrics. Our findings reveal a high classification accuracy across all models, with BERT achieving the highest score at 96.63%, indicating its superior capability in capturing contextual nuances for this mental health task. This work highlights the potential of transformer embeddings in mental health detection and provides insights into the relative strengths of each model for text classification tasks.

Index Terms— BERT, RoBERTa, DistilBERT, Electra, Depression Detection, Reddit Posts

I. INTRODUCTION

Mental health has become a pressing global concern, with depression being one of the most prevalent disorders affecting millions of individuals worldwide. Early detection of depression can significantly improve outcomes by enabling timely intervention. Social media platforms, such as Reddit, have emerged as rich sources of textual data where individuals often express their thoughts and emotions candidly. Analyzing these posts offers an opportunity to detect signs of depression using natural language processing (NLP) techniques.

In this study, we focus on leveraging state-of-the-art transformer-based language models—BERT, RoBERTa, DistilBERT, and ALBERT—to classify Reddit posts as "Depressed" or "Non-depressed." These models are known for their ability to capture deep contextual relationships within text,

making them particularly suitable for mental health classification tasks. By utilizing embeddings generated from these models and training a Logistic Regression classifier, we aim to evaluate their effectiveness in accurately identifying signs of depression. This work not only contributes to the development of robust NLP methods for mental health detection but also provides a comparative analysis of multiple transformer models in this critical domain.

II. LITERATURE SURVEY

The paper[1] underlines the importance of early intervention in depression, as delayed treatment can often lead to critical effects, even suicide, especially in teenagers and young adults. It thus highlights the weaknesses in the traditional methods of diagnosis and proposes a novel approach known as sentiment analysis—a subcategory of Natural Language Processing (NLP)—in order to ascertain the depressive tendency over the emotional communication of text, which can be gathered from multiple internet media sites and clinical files. The authors carry out a comprehensive literature review, which provides the reader with a range of methodologies and their respective efficacy in detecting depression. They come up with different sources of data: social media, online forums, personal diaries, and clinical records, and discuss the ethical implications and privacy concerns associated with such data. In this methodology, the approach is elaborated on where quantitative research describes making use of a supervisory machine learning model in training artificial intelligence models that would be able to conduct sentiment analysis and detect depression. As the approach is for descriptive and correlational analysis regarding sentiment and depression indicators, the authors explain the process of gathering data from different sources while ensuring that the data collected is relevant, representative, and

anonymized for privacy reasons. The results reveal that deep learning algorithms outperformed other sentiment analysis methods, suggesting their superior ability to handle complex language patterns and emotions. The paper also presents a correlation analysis between sentiment and depression indicators, showing that positive sentiment correlates inversely with

depressive symptoms, while negative sentiment correlates positively. In conclusion, while the paper posits the promise of sentiment analysis for early detection of depression and stresses the need to advance further research to deal with privacy over data issues, generalization of the models, and robust calibration for validation and evaluation, the authors give a comprehensive overview of AI-based detection of depression and shed light on the ethical considerations and future directions of research in this very critical area.

The paper[2] proposes a hybrid deep learning model called "Fasttext Convolution Neural Network with Long Short-Term Memory (FCL)," using techniques of natural language processing to detect depression with very high accuracy from text on so-cial media. In the proposed FCL method, combining fasttext embedding with a hybrid classifier of CNN and LSTM gains higher accuracy than other deep learning techniques such as CNN and LSTM-RNN combined with word2vec and GloVe embeddings in detecting depression from social media text. The use of fasttext embedding within the FCL method further leads to giving semantic information containing better representations of text, thus improving depression detection performance. The FCL method achieves 87 accuracy on the dataset Reddit and 88 accuracy on the dataset Twitter, better than the current methods.

The paper[3] introduces a new approach to identifying five types of depression: bipolar, atypical, psychotic, major depressive disorder, and postpartum depression from Twitter data using machine learning and deep learning models where the BERT model achieved a high accuracy of 96 and, for the first time, incorporates an explainable AI component that helps the identification of main words and phrases within the tweets relevant to the prediction behavior of the model. The BERT model achieved an overall accuracy of 0.96 in classification for predicting five types of depression: bipolar, major, psychotic, atypical, and postpartum based on Twitter data. Explainable AI highlighted which aspects of the tweets caused the depression type predictions. The study created a dataset from scratch of depression types and used lexicon labelling for predicting the type of depression.

The paper[4] proposes a deep learning-based approach to detect and evaluate the intensity of depression in Twitter users based on their tweets, using the clinical assessment criteria outlined in the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5-TR), and collaborating with expert psychologists to create a comprehensive dataset and validate the results. The proposed AI-enabled deep learning model for depression detection and evaluation achieved superior performance compared to baseline models. The study created a comprehensive dataset of 95,322 tweets labeled as "non-depressed" or "depressed" with further categorization into "mild," "moderate," and "severe" intensity levels, based on clinical assessment following the DSM-5-TR guidelines. The proposed approach combines emotional features, topical events, and behavioral-biometric signals to train the long short-term memory (LSTM)-based deep learning models for depression detection and intensity estimation. The paper discusses a multi-modal approach for the identification of depression that uses facial expression analysis, audio evaluation, and text-based analysis, also with an added intelligent chatbot that can really offer personalized support and help the system be much more accurate in identifying people at risk of depression.

The proposed system in paper[5] achieved high accuracies

during the experiments by analyzing the user's input text using facial expression analysis, audio processing, and NLP for a multi-modal depression detection system. The accuracy in the CNN model attained 92, and the accuracy in the NLP component was achieved as 92. - Intelligent chatbot integration with the system provided personalized support and encouragement to the users, which consequently improved the effectiveness of the depression detection system. Applying this multi-modal approach using facial expression, audio, and text analysis has proven to be more efficient than a single modality for depression identification.

Cherreddy et al. [6] applied NLP techniques to automatically classify tweets into categories of positive and negative sentiment categories during the process of studying community characteristics such as gender, age, and location. They used classifiers such as Naïve Bayes, Support Vector Machines (SVM), and Logistic Regression, with SVM resulting in 85% accuracy. Despite promising outcomes, the research highlighted the importance of bigger corpora in enhancing recall and precision, thus reiterating the excellent potential of NLP in supporting psychologists, policymakers, and researchers.

Jayasinghe et al. [7] proposed a system integrated real-time behavioral data with social media activity for measuring depression levels by analyzing multimodal data comprising audio, video, and text validated with the BDI-II scale with the help of CNN and NLP. The system performed high accuracy analysis in terms of subcomponents, thus proving the feasibility of integrating real-time data and social media analysis for holistic depression detection.

Athar et al. [8] proposed a Metaverse-based recommendation system based on the BERT model to identify signs of depression and anxiety. Their system not only diagnosed mental health conditions but also issued personalized recommendations for medication, psychiatrists, and therapeutic activities like music and exercise. This study elaborated on the transformative potential of immersive virtual environments in the improvement of mental health solutions using interventions tailored to individuals.

Al Asad et al. [9] aimed at doing an analysis of social media, including those such as those on the platforms of Twitter and Facebook to detect depressive symptoms. Their model using SVM and Naïve Bayes algorithms represented an effective substitute for conventional interview-based diagnostic techniques. The study processed the textual data obtained from user-generated content to demonstrate how machine learning could provide scalable solutions toward the monitoring of mental health.

Gupta et al. [10] conducted comparative analysis of six advanced NLP models, including BERT, RoBERTa, and XLNet, to conduct depression detection on Twitter. According to the authors, XLNet, DistilBERT, and RoBERTa achieved accuracy rates of more than 99% while identifying nuanced textual patterns reflecting depression. These models extensively outperformed earlier methods and were therefore highly applicable for mass-scale applications.

The paper[11] developed a depression detection system using ensemble machine learning and natural language processing techniques. The methodology involved preprocessing social

media posts by removing unnecessary symbols and stopwords, and applying stemming. Word embedding models like TF-IDF and Bi-grams were used to convert text to numerical vectors. Multiple machine learning algorithms were applied, including Logistic Regression, SVM, KNN, XGBoost, Naive Bayes and Random Forest. The best performing models were ensembled using techniques like blending, mean and median ensembling. The results showed that blending using KNN produced the best performance, achieving 96.35% accuracy, 97.08% precision, and 96.35% recall. This outperformed individual classifiers and other ensembling methods. The system demonstrated strong capability in detecting depression from social media posts, though future work could expand to multiple languages and analyze emoticons.

The paper[12] examined Reddit posts to detect depression using natural language processing (NLP) techniques and machine learning approaches. The methodology included data pre-processing of Reddit posts, including tokenization and removal of URLs/punctuation/stop words, feature extraction using LIWC dictionary, LDA topic modeling, and N-gram features and classification using logistic regression, SVM, random forest, AdaBoost, and multilayer perceptron. Best single feature was bigrams with SVM classifier, achieving 80% accuracy and 0.79 F1 score. LIWC outperformed LDA as a single feature. Best overall performance was achieved by combining LIWC+LDA+bigram features with MLP classifier, reaching 91% accuracy and 0.93 F1 score. Language predictors of depression included self-preoccupation, negative emotions, suicidal thoughts, and hopelessness. The authors conclude that proper feature selection and combinations can significantly improve depression detection performance from social media text.

The paper[13] presents a novel AI-based method for detecting depression using advanced transformer networks to analyze clinical interviews. The methodology employs BERT-based models, GPT-3.5, and ChatGPT-4 to detect depression from linguistic patterns and contextual information. The study utilizes the DAIC-WOZ and Extended-DAIC datasets, as well as simulated clinical interview data collected specifically for this research. The results show that the proposed approach significantly outperforms previous methods, with GPT-3.5 achieving the highest performance (F1 score of 0.78) on the DAIC-WOZ dataset. The BERT-based model trained on the combined DAIC-WOZ and Extended-DAIC datasets achieved an F1 score of 0.82. The authors highlight the potential of their approach in revolutionizing mental healthcare through early depression detection and intervention. They also acknowledge the ethical, legal, and social implications of using AI in mental health diagnostics.

The paper[14] presents a study on depression detection using social media data, particularly Twitter posts. The methodology involves collecting tweets and applying machine learning techniques for analysis. Specifically, the authors use Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) networks, and Natural Language Processing (NLP) to classify tweets as positive, negative or moderate. The data is pre-processed by removing special characters and emojis before applying the machine learning models. The results show that LSTM performs better than RNN, with 89% accuracy compared

to 70% for RNN. However, a hybrid LSTM+RNN model achieved the highest accuracy of 91%. The authors conclude that machine learning and NLP techniques can effectively detect depressed tweets, which could help identify users at risk of depression or suicide. They suggest future work to develop systems that can automatically delete or filter out depressive content on social media platforms.

The study[15] employed natural language processing (NLP) and machine learning (ML) techniques to detect depression from social media text. The methodology involved data preprocessing using NLP lemmatization and stemming, followed by feature extraction using CountVectorizer, N-gram analysis, and TF-IDF. Seven ML classifiers were utilized: Naive Bayes, Stochastic Gradient Descent, Support Vector Machine, Decision Tree, Random Forest, Multi-layer Perceptron, and K-Nearest Neighbor. Results showed that for Dataset-1, SVM achieved the highest accuracy at 97.00%, followed by Random Forest (91.00%), Multi-Layer Perceptron (90.00%), and Decision Tree (88.00%). For Dataset-2, Random Forest performed best with 94.28% accuracy, followed by Multi-Layer Perceptron (93.00%), Stochastic Gradient Descent (92.00%), and SVM (88.00%). The Support Vector Machine emerged as the top-performing classifier across both datasets. The model was exported and integrated into a browser-based interface using Streamlit for practical application and user testing.

In the study [16] by Bhavya R., et. al. the authors apply machine learning methods for identifying depression in Social Media Posts. This study analyzes a dataset of 20,000 English tweets through sentiment analysis and classification algorithms such as Decision Trees, Random Forest, K-Nearest Neighbours (KNN) and Naive Bayes, showing the effectiveness of these methods in predicting the content of depression. Among the tested models, the decision tree algorithm achieved the highest accuracy with 93.97%, followed by the random forest classifier with 92.95%. Naive Bayes and KNN showed lower accuracy rates, indicating the different effectiveness of the different algorithms in the classification of sentiment. This study demonstrates the potential of social media data as a useful resource for the early diagnosis of depression and highlights the importance of addressing challenges such as overfitting in decision trees and assumption of independence in Naive Bayes. The authors call for further research to confirm these findings in larger datasets, and to examine the ethical implications of using artificial intelligence for mental health diagnosis.

The research article [17] by Sadeghi et al. investigates the use of large linguistic models (LLM) to detect depression through text analysis, specifically using E-DAIC data. The study ignores traditional methods of manual feature extraction, instead using an automated process via LLM to predict PHQ-8 scores, which measure the severity of depression. By fine-tuning the DepRoBERTa model on processed text data using GPT-3.5-Turbo queries, the researchers obtained a mean absolute error (MAE) of 3.65, demonstrating the potential of LLM to improve depression screening accuracy. This work contributes to the growing evidence that advanced NLP techniques can significantly improve mental health assessments by automatically extracting subtle linguistic features indicative of depression, providing a scalable and effective approach to diagnosis and early intervention.

In the study [18], Dessai, S., et al. study the application of text mining and natural language processing (NLP) techniques to detect depression through tweets. Using a Convolutional Neural Network (CNN) combined with Long Short-Term Memory (LSTM) networks, the authors achieved a remarkable accuracy of 92% in identifying depressive content, outperforming traditional methods such as Logistic Regression and Term Frequency-Inverse Document Frequency (TF-IDF). Their research highlights the effectiveness of advanced neural network architectures in capturing subtle language patterns associated with depression, reflecting the potential of social media data for mental health monitoring. The study highlights the importance of text mining for extracting information from unstructured social media content and suggests future research directions to refine detection models and explore the impact of demographics of users and personality traits on mental health detection.

The study [19] by Zaman, A., et al. presents a new approach for depression detection using a refined RoBERTa model applied to Twitter posts. This research focuses on harnessing social media data to improve early diagnosis and preventive care. The authors propose a model that refines RoBERTa, a powerful transformer-based linguistic model, to classify depression levels with a high degree of accuracy. Their model achieved a remarkable 90% accuracy in predicting the severity of depression, outperforming many existing approaches that often focus on binary classifications or less nuanced methods. The study includes advanced sentence-level contextual analysis, which improves the model's ability to interpret the subtleties of social media text better than traditional word-level methods. Previous research has highlighted the potential of social media for detecting mental health, with different models achieving different levels of accuracy. However, the focus on multilevel classification in this study fills a critical gap by providing a more granular understanding of depression, which can significantly improve clinical outcomes. Overall, this work provides valuable insights into the application of advanced machine learning techniques to mental health monitoring and presents a solid framework for future research and clinical practice.

The study [20], authored by Sanga, P., et al., explores the integration of attention mechanisms in deep learning models to improve depression detection from textual data. With depression rates rising worldwide, traditional sentiment analysis methods struggle to accurately distinguish depression from text data, often resulting in insufficient detection capabilities. This research addresses these limitations by comparing five deep learning models with attentional mechanisms to their non-attention counterparts. The study used two datasets, Reddit and Kaggle, to validate the performance of these models. The results show a significant improvement in the accuracy and robustness of the attention-driven models, with an average accuracy increase of 1.52% on the Reddit data and 1.21% on the entire Kaggle data. In particular, the DistilBERT-BiLSTM model achieved the highest accuracy of 94.93% with an AUC score of 0.9501, indicating its superior performance. This study confirms that attentional mechanisms significantly improve the robustness and efficiency of depression detection models, offering a promising path for future research in mental health analysis.

III. METHODOLOGY

A. DistilBert

Utilizing DistilBERT, a lightweight transformer model, to make embeddings out of cleaned Reddit posts for depression classification. It has been chosen due to its efficiency: It maintains 97% of BERT's performance and makes it 40 smaller and 60% faster; hence, this model is suitable for resource - constrained tasks. The dataset is loaded and preprocessed, with texts tokenized using the DistilBERT tokenizer with truncation and padding to get up to a maximum sequence length of 128 tokens. The model DistilBERT will generate contextual embeddings for each post using the last hidden state averaged to produce a fixed-size representation. For a logistic regression

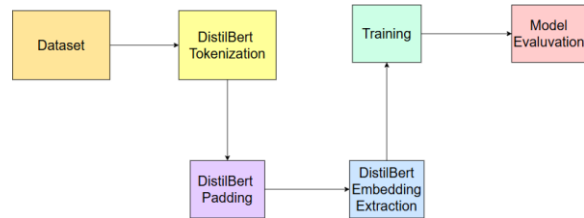


Fig. 1. DistilBERT Architecture

model, these embeddings can be considered input features. Class weights will be computed using 'compute_class_weight' and incorporated into the logistic regression classifier in order to deal with the problem of class imbalance. Then the model trains and evaluates the model, printing accuracy alongside a verbose classification report to see how well it's doing in distinguishing between the "Depressed" and the "Non-depressed" posts. DistilBERT is computationally efficient with very strong contextual understanding and thus makes up for a very robust choice for the classification task.

B. Roberta

Our approach utilized a preprocessed dataset containing Reddit posts, each labeled as either "Depressed" or "Non-depressed." We leveraged RoBERTa, a transformer-based language model, to extract meaningful semantic embeddings for text classification. Specifically, we initialized the RobertaTokenizer and RobertaModel from the roberta-base pre-trained model. For each post, tokens were generated using the tokenizer with truncation, padding, and a maximum sequence length of 128. The RoBERTa model then produced contextualized token embeddings, from which we computed sentence-level embeddings by averaging the last hidden layer's token outputs. These embeddings were detached and stored as NumPy arrays. To handle potential class imbalance, we calculated class weights based on the training labels and integrated these weights into the Logistic Regression classifier. The data was split into training and testing subsets using an 80:20 ratio, ensuring a balanced representation of both classes. The Logistic Regression model was then trained with a maximum of 1000 iterations on the extracted embeddings, followed by predictions on the test set. To evaluate model performance, we calculated metrics including accuracy, precision, recall, and F1-scores, with a detailed classification report generated for both classes.

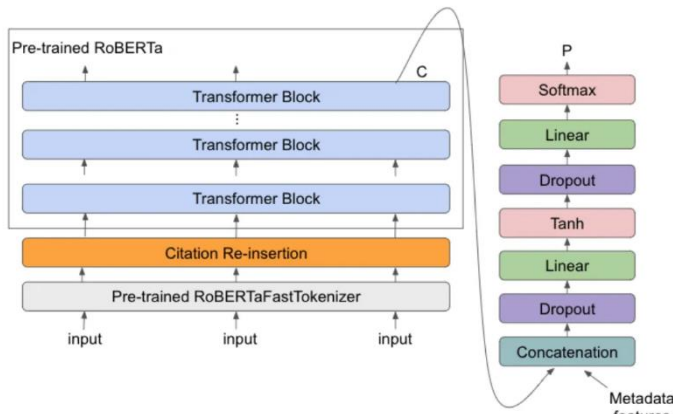


Fig. 2. Roberta Architecture

C. Electra

The methodology starts with the loading and preprocessing of the dataset. Importing pandas imports cleaned Reddit posts with corresponding labels that indicate whether the post is related to depression. The text data is extracted from the 'clean_text' column while the labels are obtained from 'is_depression'. The data shall then be preprocessed with text tokenization, using the ELECTRA tokenizer, to ensure that all input sequences are truncated to a maximum length of 128 tokens and padded to ensure that all sequences are uniform in length. This needs to be done ahead of arranging the data in a suitable format for model input to be further processed.

Upon preparing the data, the ELECTRA model was used to produce contextual embeddings for each Reddit post. This tokenized input text was passed through the ELECTRA model which would result in a sequence of hidden states at every layer of the transformer. The last hidden state-based embeddings were obtained and then averaged over all tokens in the sequence to encapsulate the essence of the post. The averaging process generates a fixed-size vector representation for each post. This is supposed to effectively summarize the content of the post while preserving contextual information. These embeddings serve as input features for classification tasks.

The dataset is then split into two: the training set and the test set, using 80:20. That is to say 80% will be used for training to the model, leaving 20% for testing. This way, the model will have enough information it can use in learning, but at the same time provide a separate, unseen test dataset to test the performance of the model. To deal with potential class imbalance, in which one class (very likely the "Non-depressed") would dominate, class weights will be calculated using sklearn's `compute_class_weight` function. Class weights will shift the importance of each class, that is to say allow the model to learn from a minority more effectively ("Depressed").

The final task of classification is completed using a Logistic Regression classifier. The classifier is trained on the embeddings produced by the ELECTRA model. Class weights are employed in the logistic regression model, such that the model adequately compensates for the imbalance present in the data set. By training on those embeddings and modified class weights, the logistic regression model should be able to make a distinction between "Depressed" and "Non-depressed" posts using the rich contextual information contained in the text.

The test set is used to evaluate the performance of the model by generating predictions for the test set and then calculating the accuracy of the model. In addition to accuracy, a classification report is created that provides a clear definition of the model's performance through precision, recall, and F1 score for each respective class. These will provide the measures necessary to understand the capability of the model in terms of its ability to effectively tag posts as depression and other posts as well.

Choosing ELECTRA for the task was on the basis of its strong contextual understanding and computational efficiency. Unlike other models like BERT, ELECTRA shows comparable performance but greater speed and resources. This makes ELECTRA an ideal choice when computational resources are limited but high performance is required. Bringing logistic regression along with ELECTRA brings out a method, which can represent an efficient solution to classify Reddit posts as "Depressed" or "Non-depressed".

D. Bert

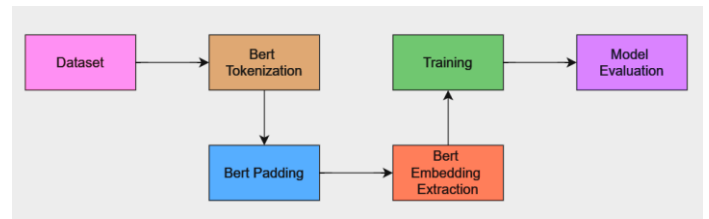


Fig. 3. Bert Architecture

This methodology incorporates several key steps: data preparation, feature extraction using a pre-trained BERT model, model training, and finally, evaluation. It effectively combines the contextual capabilities of BERT with the simplicity and efficiency of logistic regression for binary classification. The methodology leverages both high-quality dataset provision and the advancements of state-of-the-art NLP techniques to deliver accurate and interpretable predictions in the identification of signs of depression in text data.

Data Preprocessing is the first step of the procedure. The dataset used for this experiment is presented in a tabular manner with the inclusion of preprocessed text data and their corresponding binary labels representing the absence or presence of depression. Here, the text data represent an input to the feature extraction, and the labels act as ground truth for the classification process. In guaranteeing the presentability of the extracted data in an appropriate format for further processing, extracting the relevant columns from the dataset is one thing; proper cleaning and preprocessing of data are necessary as shown in pre-cleaned text data to ensure the next stages of the pipeline focus on content relevant to the text.

After the data preparation step, feature extraction through a pre-trained BERT model is performed on the text. BERT is short for Bidirectional Encoder Representations from Transformers. It is a deep learning-based model intended to understand the contextual meaning of words in a given sentence. The usage of BERT has been pivotal in this approach because it discovers subtle relationships and semantic meanings that traditional word embeddings often miss. Preprocessing involves tokenizing the input text using a BERT pretrained tokenizer which reduces it into a format digestible by the model. It pads text and truncates text to ensure consistent length input across stages. This process has a given weight as an important

parameter in improving computer efficiency and ensuring uniformity among the text samples used.

After tokenization the text is passed through a pretrained BERT model in order to yield contextual embeddings for each token. These embeddings encapsulate the semantics and syntax of the wider context of the text. The embeddings across all the tokens in a text sample are aggregated using mean pooling, such that each text sample will be represented as a single feature vector. This will then give a compact but rich text representation that may be used as input for the classification model. The capacity to produce highly deep contextualized representations makes it very effective for tasks such as depression detection, where subtle differences of language use can indicate the status of mental health.

Having prepared the feature vectors, it's time to split the data into training and testing sets. So that a big portion of the data is kept for training while another portion is kept for evaluation. This makes sure it is learned by one of the subsets of data and its generalization ability is tested on an unseen subset. The division is performed in that way that maintains a balance between the classes so that equal proportions of both depressed and non-depressed samples will be represented in both sets. Reproducibility is ensured by generating a static random state in the course of partitioning.

The classification model applied in this methodology is logistic regression. Logistic regression is another light-weight and interpretable machine learning algorithm that is particularly well suited to binary classification tasks. Despite how trivial this sounds, with really good features, this is basically one of the most effective algorithms to use. The Model The model is trained from a set of feature vectors drawn from the training set, along with their corresponding labels. Training involves optimizing model parameters to best minimize classification error. It is chosen because logistic regression can handle data with high dimension very well, and arguably it is less likely overfitting against more complex models.

After the training phase, the logistic regression model is assessed on the test set. Assessment involves creating predictions for test samples and comparing such predictions with labels of the actual instances. The main accuracy measure that is used to test the performance of the model is the accuracy, in terms of instances that are correctly classified. Moreover, it provides a classification report that consists of its respective precision, recall, and F1 score for each class. It gives an exhaustive understanding of where the model is going wrong, particularly about depressing and non-depressing samples. In general, such a methodology combines the strengths of BERT's deep contextual embeddings with the simplicity of logistic regression, thus creating an efficient framework for the detection of depression from textual data. BERT embeddings handle all the subtleties and complexities of natural language so deeply implicit in capturing subtle indicators of depression. Logistic regression ensures that the system remains computationally efficient and interpretable and hence suitable for deployment in real-world scenarios. Such methodology would imply advanced natural language processing with standard machine-learning techniques so that the solution for detecting depression is scalable and very effective.

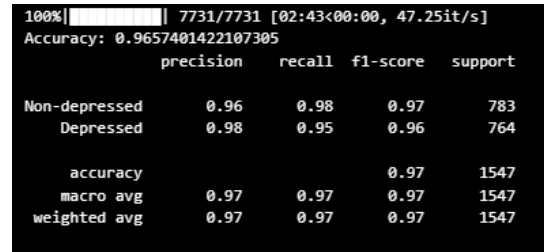
E. Prompt Engineering

The methodology for the depression detection system starts from the user-input text, wherein a system message is configured to send the input into the system for classification. The initial phase would be to set up a system message to tell the AI model to classify the given text as either "depressed" or "not depressed." Appended to the session as a human message, the input from the user sends it to the OpenAI model via the LangChain framework. The model will receive the input and pass a response, which will be stored as an AI message in the session state. Finally, the classification result will be presented to the user to offer feedback on the entered text and help identify potential signs of depression.

IV. RESULTS

A. DistilBert

With the test set accuracy at an impressive 96.57%, it is very strong in classifying the posts about depression on Reddit. Precision, recall, and F1-score have been consistent, always on or around 0.97, in both the macro and weighted classification. This would mean that the model is doing very well for detection and also in the correct identification of the class concerning cases of depression, while at the same time balancing both classes. The results show the effectiveness of using the DistilBERT embeddings with logistic regression for this particular classification task.



	precision	recall	f1-score	support
Non-depressed	0.96	0.98	0.97	783
Depressed	0.98	0.95	0.96	764
accuracy			0.97	1547
macro avg	0.97	0.97	0.97	1547
weighted avg	0.97	0.97	0.97	1547

Fig 4:Classification Report DistilBert

B. Roberta

Our model achieved an impressive accuracy of **96.44%** on the test set, highlighting its effectiveness in identifying mental health indicators from Reddit posts. The classification metrics showed a precision of **0.95**, recall of **0.98**, and an F1-score of **0.97** for the "Non-depressed" class. Similarly, the "Depressed" class achieved a precision of **0.98**, recall of **0.95**, and an F1-score of **0.96**, demonstrating the model's balanced performance. The macro-average and weighted-average F1-scores were both **0.96**, further confirming that our approach effectively captured nuanced differences between the two classes. These results underscore the potential of RoBERTa-based embeddings combined with Logistic Regression for reliable and accurate mental health classification.

Accuracy: 0.9644473173884939				
	precision	recall	f1-score	support
Non-depressed	0.95	0.98	0.97	783
Depressed	0.98	0.95	0.96	764
accuracy			0.96	1547
macro avg	0.96	0.96	0.96	1547
weighted avg	0.96	0.96	0.96	1547

Fig 5: Classification Report for Roberta

C. Electra

Our model achieved an impressive accuracy of 93.79% on the test set, demonstrating its effectiveness in identifying mental health indicators from Reddit posts. The classification metrics for the "Non-depressed" class showed a precision of 0.93, recall of 0.95, and an F1-score of 0.94, while the "Depressed" class achieved a precision of 0.94, recall of 0.93, and an F1-score of 0.94. The macro-average and weighted-average F1-scores were both 0.94, highlighting the model's balanced performance across both classes. These results showcase the potential of ELECTRA-based embeddings combined with Logistic Regression for reliable and accurate depression classification in natural language text.

	precision	recall	f1-score	support
Non-depressed	0.93	0.95	0.94	783
Depressed	0.94	0.93	0.94	764
accuracy			0.94	1547
macro avg	0.94	0.94	0.94	1547
weighted avg	0.94	0.94	0.94	1547

Fig 6: Classification Report Electra

D. Bert

The proposed method was found to work well in the classification of depression with an accuracy of 96.51%. The precision, recall and F1-scores for classes "Non-depressed" and "Depressed" were not biased differently and proved to be flat. The precision of class "Non-depressed" was 0.95, the recall was 0.98 and the F1-score was 0.97. The precision of class "Depressed" was 0.98, the recall was 0.95 and the F1-score was 0.96. The macro and weighted averages for these metrics further establish that there is more consistency in different classes. These results highlight the effectiveness of using BERT embeddings in combination with logistic regression for accurate mental health classification.

Accuracy: 0.9650937297996122				
	precision	recall	f1-score	support
Non-depressed	0.95	0.98	0.97	783
Depressed	0.98	0.95	0.96	764
accuracy			0.97	1547
macro avg	0.97	0.96	0.97	1547
weighted avg	0.97	0.97	0.97	1547

Fig. 7. Classification Report Bert

E. Prompt Engineering

The result of the depression detection system will classify the user's input as "depressed" or "not depressed" according to what the model deems appropriate. Streamlit will then output this classification result to the user in the form of providing him with immediate feedback on his text.

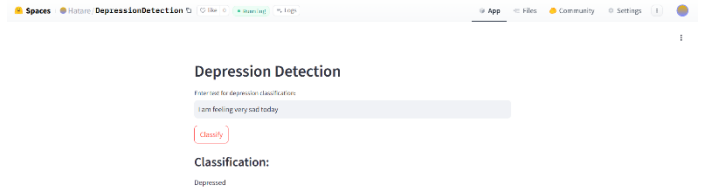


Fig 7: Depressed Class

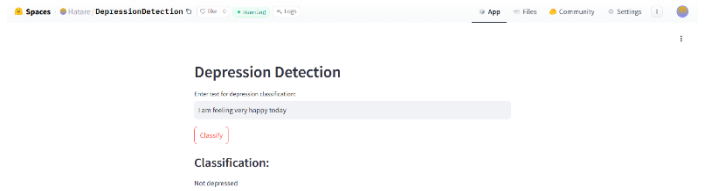


Fig 8: Not Depressed Class

REFERENCES

- [1] Choudhary, D.K., Gupta, A., Singh, S., Abhinav, T. and Agrawal, T., 2024, May. Sentiment Analysis for Depression Detection Using Artificial Intelligence. In 2024 3rd International Conference on Artificial Intelligence For Internet of Things (AIIoT) (pp. 1-5). IEEE.
- [2] Tejaswini, V., Sathya Babu, K. and Sahoo, B., 2024. Depression detection from social media text analysis using natural language processing techniques and hybrid deep learning model. ACM Transactions on Asian and Low-Resource Language Information Processing, 23(1), pp.1-20.
- [3] Nusrat, M.O., Shahzad, W. and Jamal, S.A., 2024. Multi Class Depression Detection Through Tweets using Artificial Intelligence. arXiv preprint arXiv:2404.13104.
- [4] Prama, T.T., Islam, M.S., Anwar, M.M. and Jahan, I., 2024. AI-Enabled Deep Depression Detection and Evaluation Informed by DSM-5-TR. IEEE Transactions on Computational Social Systems.
- [5] Sharma, A., Saxena, A., Kumar, A. and Singh, D., 2024, March. Depression Detection Using Multimodal Analysis with Chatbot Support. In 2024 2nd International Conference on Disruptive Technologies (ICDT) (pp. 328-334). IEEE.
- [6] Chereddy, S., Geetha, K., & Sreedevi, A. G. (2024, May). Tweeting the Blues: Leveraging NLP and Classification Models for Depression Detection. In 2024 2nd International Conference on Advancement in Computation & Computer Technologies (InCACCT) (pp. 875-880). IEEE.
- [7] Jayasinghe, G. C. J., Shamika, I. P. M. A., Dissanayake, G. A. I. P., Ranaweera, R. M. I. A., & Bandara, P. S. (2022, December). Depression Detection System Using Real-Time and Social Media Data. In 2022 4th International Conference on Advancements in Computing (ICAC) (pp. 168-173). IEEE.
- [8] Athar, A., Mozumder, M. A. I., Fathima, K., Hussain, A., Ali, S., & Kim, H. C. (2023, September). Anxiety and Depression Detection and Rehabilitation in the Metaverse: A BERT-Based Recommendation System. In 2023 International Conference on Intelligent Metaverse Technologies & Applications (iMETA) (pp. 1-5). IEEE.
- [9] Al Asad, N., Pranto, M. A. M., Afreen, S., & Islam, M. M. (2019, November). Depression detection by analyzing social media posts of user. In 2019 IEEE international conference on signal processing, information, communication & systems (SPICSCON) (pp. 13-17). IEEE.
- [10] Gupta, K., Jinad, R., & Liu, Q. (2023, June). Comparative analysis of nlp models for detecting depression on twitter. In 2023 International Conference on Communications, Computing and Artificial Intelligence (CCCAI) (pp. 23-28). IEEE.
- [11] Jagtap, H. Shukla, V. Shinde, S. Desai And V. Kulkarni, "Use Of Ensemble Machine Learning To Detect Depression In Social Media Posts," 2021 Second International Conference On Electronics And Sustainable Communication Systems (Icesc), Coimbatore, India, 2021, Pp. 1396-1400, Doi: 10.1109/Icesc51422.2021.9532838.
- [12] M. M. Tadesse, H. Lin, B. Xu and L. Yang, "Detection of Depression-Related Posts in Reddit Social Media Forum," in IEEE Access, vol. 7, pp. 44883-44893, 2019, doi: 10.1109/ACCESS.2019.2909180.
- [13] M. Danner et al., "Advancing Mental Health Diagnostics: GPT-Based Method for Depression Detection," 2023 62nd Annual Conference of

- the Society of Instrument and Control Engineers (SICE), Tsu, Japan, 2023, pp. 1290-1296, doi: 10.23919/SICE59929.2023.10354236.
- [14] S. S. Korti and S. G. Kanakaraddi, "Depression detection from Twitter posts using NLP and Machine learning techniques," 2022 Fourth International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT), Mandya, India, 2022, pp. 1-6, doi: 10.1109/ICERECT56837.2022.10059773.
 - [15] F. Kabir, M. A. Hossain, A. F. M. Minhazur Rahman and S. Z. Mishu, "Depression Detection From Social Media Textual Data Using Natural Language Processing and Machine Learning Techniques," 2023 26th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2023, pp. 1-6, doi: 10.1109/ICCIT60459.2023.10441612.
 - [16] Bhavya, R., Ramani, R., Royal, G. L. G., Manideep, G., & Sridhar, G. (2024, April). Exploring Depression Through Social Media: A Textual Analysis. In 2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS) (Vol. 1, pp. 1-7). IEEE.
 - [17] Sadeghi, M., Egger, B., Agahi, R., Richer, R., Capito, K., Rupp, L. H., ... & Eskofier, B. M. (2023, October). Exploring the capabilities of a language model-only approach for depression detection in text data. In 2023 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI) (pp. 1-5). IEEE.
 - [18] Dessai, S., & Usgaonkar, S. S. (2022, May). Depression detection on social media using text mining. In 2022 3rd International Conference for Emerging Technology (INCET) (pp. 1-4). IEEE.
 - [19] Zaman, A., Ferdous, S. S., Akhter, N., Ena, T. I., Nabi, M. M., & Asma, S. A. (2023, September). A Multilevel Depression Detection from Twitter using Fine-Tuned RoBERTa. In 2023 International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD) (pp. 280-284). IEEE.
 - [20] Sanga, P., Singh, J., & Priyadarshi, P. (2023, November). Unmasking Depression via Attention-modulated Text Analysis using Deep Learning. In 2023 International Conference on Communication, Security and Artificial Intelligence (ICCSAI) (pp. 335-340). IEEE.

